DINSAR DATA INTEGRATION IN VULNERABILITY ANALYSES OF BUILDINGS EXPOSED TO SLOW-MOVING LANDSLIDES

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ABSTRACT

High and very-high resolution remote sensing data provided by the processing of Synthetic Aperture Radar (SAR) images via Differential Interferometry techniques (DInSAR) allow detecting ground surface displacements that can be used in monitoring/control of urban areas and integrated within procedures aimed at analyzing the vulnerability of affected facilities (e.g. structures and infrastructure). In this regard, this paper presents the results of a study that combines DInSAR data and information gathered from damage surveys to buildings interacting with slow-moving landslides. In particular, relationships between cause (DInSAR-derived differential settlements of buildings) and effect (recorded building damages) are retrieved and empirical fragility and vulnerability curves (to be used for damage forecasting purposes) are generated. Considering the encouraging outcomes, the adopted methodological approach can be successfully integrated within a more general framework for the management of slow-moving landslide risk to which urban areas are exposed.

Index Terms— DInSAR, monitoring of urban areas, slow-moving landslides, fragility and vulnerability curves

1. INTRODUCTION

Landslides slow-moving on preexisting sliding surfaces are widespread and their interaction with the urban environment is often responsible of detrimental consequences to structures (e.g., buildings) and infrastructure (e.g., roads, embankments, subsurface utilities). Indeed, the interaction of the above landslides with facilities not properly designed to withstand additional stresses induced by foundation movements, generally induces direct (i.e. damage) and indirect (in terms of economic losses and inconvenience to involved stakeholders) consequences. For these reasons, landslide hazard and vulnerability analyses represent the key steps for the damage prediction as well as for the proper design and implementation of the most suitable risk mitigation measures. However, the achievement of these goals usually requires high costs due to the great amount of data to be collected for setting up reliable forecasting models. These data deal with relevant information on slow-moving landslide features (e.g. volume, velocity, etc.), the exposed facilities (stiffness/strength of constituting materials, state of maintenance, value) and their damage severity level (generally associated with the attainment of certain serviceability/ultimate limit states). However, these data collection is not an easy task due to the existence of several sources of uncertainties related to *i)* the peculiarities of factors that predispose/trigger a given slow-moving landslide; *ii)* the spatial distribution of the intensity parameter and its temporal variability; *iii)* the vulnerability that might change from one facility to another even for similar slow-moving landslide intensity [1], [2].

As for the buildings interacting with slow-moving landslides, their vulnerability can be conveniently analyzed by retrieving the relationships between the cause (e.g. the magnitude of the differential settlement experienced by the foundation system) and related effect (damage severity level reached by the superstructure). The latter depend on the landslide mechanism and related intensity (i.e., differential displacements and/or their velocity whose spatial attributes result, for instance, from the position of the exposed elements with respect to the landslide affected area) as well as the building/foundation characteristics and can be evaluated based on the data collected by in situ damage surveys via the analysis of crack patterns exhibited by building façades [3], [4].

As for the intensity parameter, referring to displacement patterns (and rates) experienced by single buildings located in (or in the proximity of) slow-moving landslide-affected areas, Synthetic Aperture Radar (SAR) data processed via Differential Interferometry (DInSAR) and tomography techniques [5], can be extremely useful in providing long-term ground/building displacement archives. These well-established, non-invasive and non-destructive techniques, take advantage of *i)* the availability of past and current generation SAR sensors that offer a wide range in terms of spatial resolution, coverage and revisiting times, *ii)* several

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processing algorithms (well-described and tested by specialized research teams and private companies), *iii)* high cost-effectiveness in terms of coverage and accuracy.

In particular, most recent very high-resolution SAR sensors led to a major step towards the application of these data for the investigation of single structure/infrastructure behavior and relative causes of their failures [1], [4], [6], [7], [8], [9].

The present study shows how the combined use of DInSAR data with the results of a building damage survey can provide a useful contribution in vulnerability analysis of buildings exposed to slow-moving landslides.

The analysis was carried out with reference to the urban area of Verbicaro municipality affected by slow-moving landslides located in the northwest sector of Calabria region in the southern Italian Apennines.

2. STUDY AREA

The urban area of Verbicaro (Fig. 1) has been suffering from slope instability for a long time with several slow-moving landslides of different type [10] that interact with masonry – mainly located in the historic centre – and reinforced concrete buildings, causing damages of different severity levels [4].

For the whole study area a building damage survey carried out in April–May 2014 is available. The recorded information on the building damage – collected via ad hoc predisposed fact-sheets [3], [4] – was classified based on the interpretation of crack patterns adopting the ranking proposed by [11]. In particular, fact-sheets were filled in for 253 reinforced concrete and 239 masonry surveyed buildings and the damage severity levels were distinguished in six classes (D0 = negligible; D1 = very slight; D2 = slight; D3 = moderate; D4 = severe; D5 = very severe).

The collected data (Fig. 1) highlight a widespread state of damage that affects both masonry buildings (50% out of the total exhibit some damage level whose severity is higher than D0) and reinforced concrete structures (34% out of the total surveyed reinforced concrete buildings), these latter mainly located in a newly developed area built-up starting from the early 60s.

The used SAR dataset consists of ENVISAT (acquired on descending orbit from August 2003 to January 2010) and COSMO-SkyMed (acquired on descending orbit from April 2013 to December 2013) images, that were processed according to the SAR tomographic analysis [5], providing a high quality of information on coherent targets.

The derived DInSAR velocity distribution are show in Figure 2a and Figure 2b, respectively, for ENVISAT and COSMO-SkyMed datasets in which it can be appreciated the significant coverage increase provided by COSMO-SkyMed data, thus highlighting their effectiveness in monitoring and controlling the built-up environment.

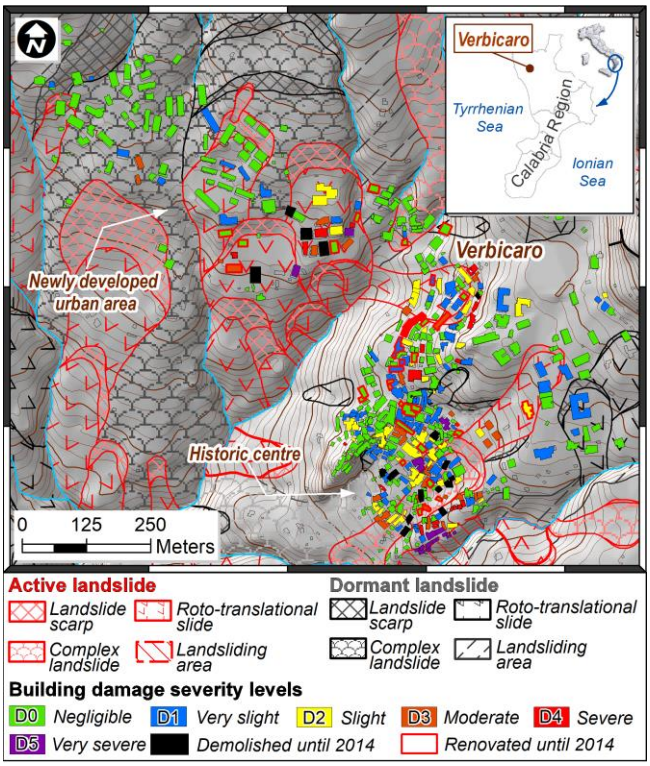


Fig. 1. The Verbicaro study area with slow-moving landslide inventory map [10] and surveyed buildings distinguished according to the recorded damage severity levels.

3. RESULTS

The analysis focused on a sub-sample of 42 masonry buildings located on landslide-affected areas that, from the results of the extensive in-situ damage survey, show damage severity levels covering all classes of the adopted ranking (from D0 to D5) and are covered by at least two coherent pixels in both ENVISAT and COSMO-SkyMed datasets. For each of the above selected buildings, the DInSAR-derived differential settlement were evaluated (Fig. 3a).

In particular, the differential settlements values were computed as the maximum difference of the cumulative settlements recorded by the pertaining coherent pixels. The cumulative settlements were obtained by multiplying the average velocity – extracted along the vertical direction (Fig. 3a) – for the period of observation of each available dataset and considering, for the period February 2010-March 2013 when DInSAR data are lacking, a constant velocity value equal to the one associated with the longest available dataset (i.e. ENVISAT) [1], [2], [4].

The combination of DInSAR-derived differential settlements with the corresponding in-situ recorded building damage severity levels, provide a preliminary relationship (Fig. 3b) between cause (differential settlement Δ) and effect (damage severity level) for masonry buildings affected by slow-moving landslides within Verbicaro urban area.

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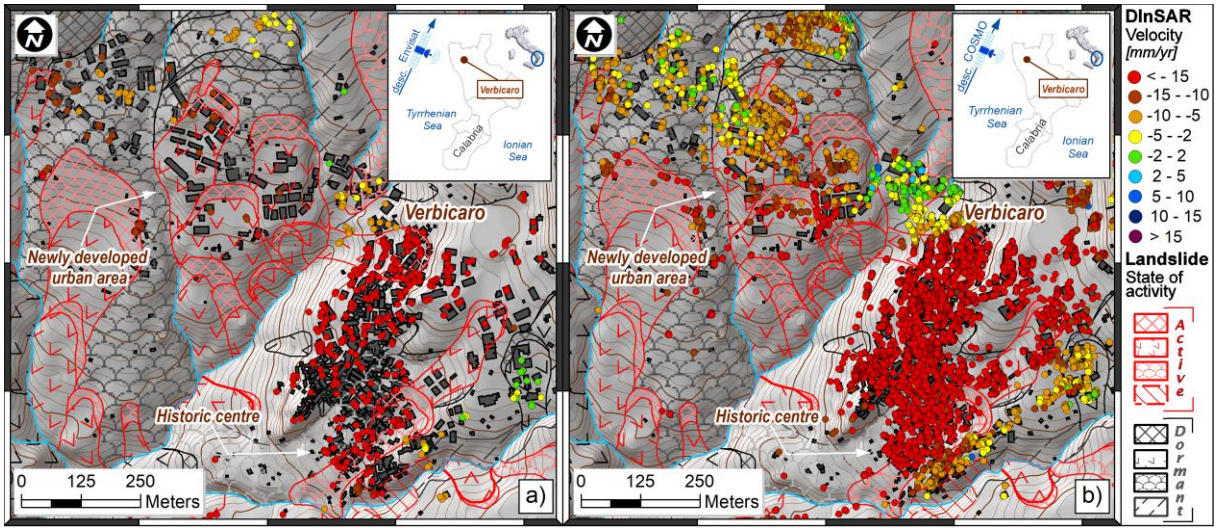


Fig. 2. DInSAR velocity distribution provided by a) ENVISAT and b) COSMO-SkyMed radar sensors on descending orbit overlaid on the slow-moving landslide inventory map.

The obtained relationship highlights a generally increasing trend of the damage severity level recorded to buildings when the representative intensity parameter (i.e. the differential settlement derived from DInSAR data), on average, increases.

Starting from the obtained diagram (Fig. 3b), empirical fragility curves (Fig. 3c), which provide the conditional probability $P(\cdot)$ for a randomly selected building at risk to be in, or exceed, a certain damage severity level (Di) when the intensity parameter (Δ) equals a given value, were generated. To this aim, a log-normal distribution function $\phi[_]$ was used as probabilistic model [1], [4], [9], [12]; whereas the fragility parameters (median α and standard deviation β) were computed using the maximum likelihood method adopting a standard optimization algorithm.

Finally, an empirical vulnerability curve (Fig. 3d), relating the expected mean level of damage severity ($\mu_D(\Delta)$) to a given building and the value of the landslide intensity parameter (Δ) was derived by fitting the $\mu_D(\Delta)$ data - obtained by multiplying the discrete probability P_i associated with each damage severity level Di (extracted from the generated fragility curves – Fig. 3c) for a numerical index d_i (taken for this application as 1, 2, 3, 4, and 5 for D1, D2, D3, D4, and D5, respectively) - using as regression model the tangent hyperbolic function [1].

4. CONCLUSIONS

The paper presented some results of a wider research focused on the integrated use of DInSAR and building damage survey data in a slow-moving landslide-affected urban areas. The achieved outcomes in terms of cause-effect relationships and empirical fragility/vulnerability curves, highlight the potential of using such a widespread information as DInSAR data for the generation of building damage forecasting models. These latter, once further calibrated and validated – in municipalities presenting similar landslide types and urban fabric – can be valuably used within procedures aimed at investigating and predicting building vulnerability in slow-moving landslide-affected urban areas as well as in the decisional processes for the selection of the most appropriate risk mitigation measures.

ACKNOWLEDGMENTS

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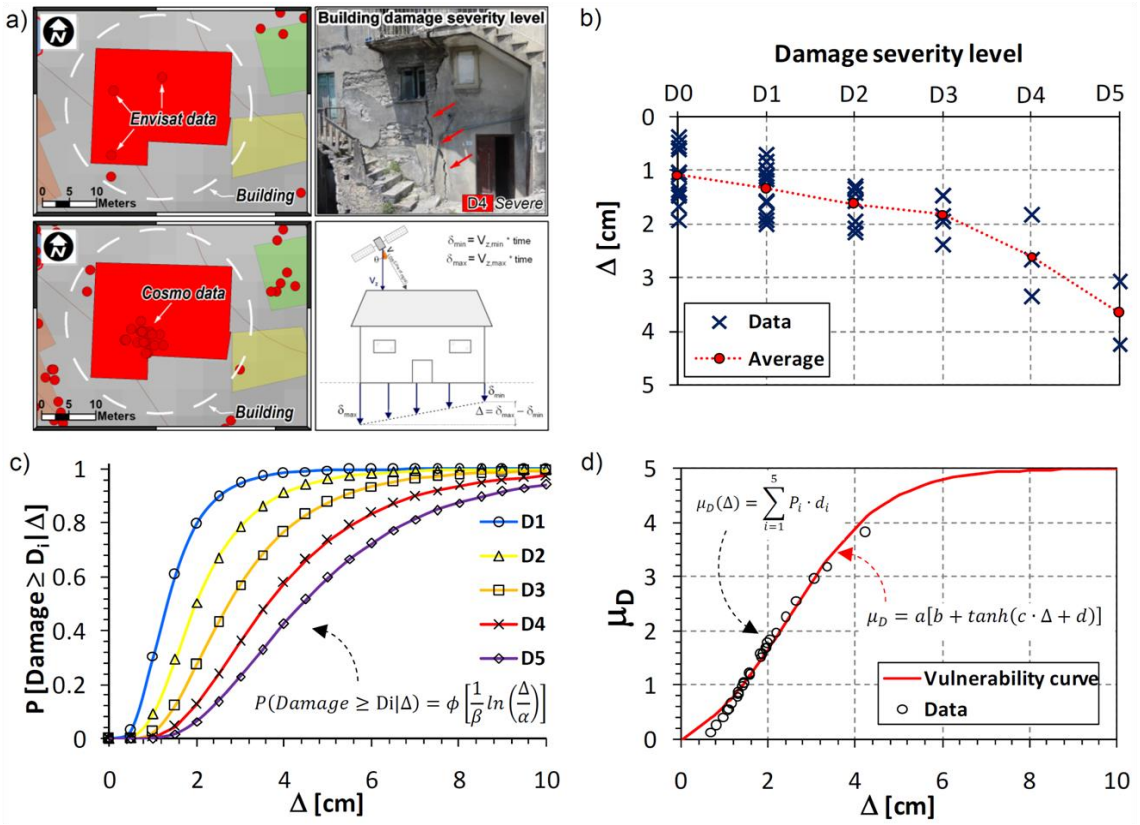


Fig. 3. Results of the analysis on the sub-sample of 42 masonry buildings in slow-moving landslide-affected areas of Verbicaro municipality: a) an example of DInSAR-derived differential settlement (Δ) for each surveyed building and recorded damage severity level; b) damage severity levels vs. DInSAR-derived differential settlements (Δ); empirical fragility c) and vulnerability d) curves.

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